

1 Empirical Results

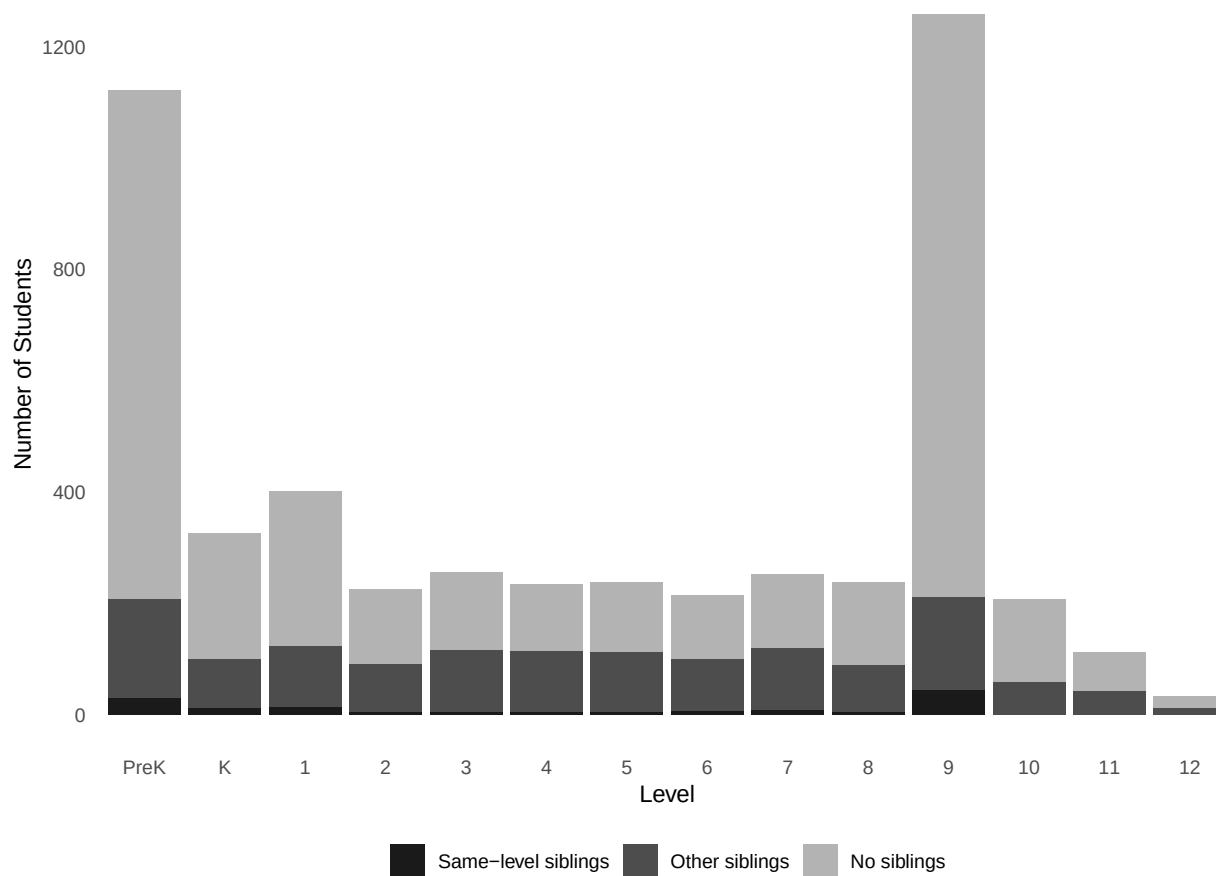


Figure 1: Summary Statistics by Region, Admissions Process 2023-2024

Table 1: Summary Statistics by Region, Admissions Process 2023-2024

Region	Schools		Students		Siblings				
					Pairs	Same Level		Block	
	N	Seats	N	Applications	N	N	%	N	%
Tarapaca	1345	18653	16577	56613	3929	189	0.048	2005	0.510
Antofagasta	1477	28428	24335	86855	4299	283	0.066	2138	0.497
Atacama	1189	20603	11910	37910	1991	143	0.072	899	0.452
Coquimbo	4800	54033	27892	101412	5345	347	0.065	2934	0.549
Valparaíso	8476	111273	52747	170340	9202	660	0.072	5340	0.580
OHiggins	4908	67203	30752	93230	4592	407	0.089	2387	0.520
Maule	6281	80627	35607	116467	5340	438	0.082	2636	0.494
Biobio	7466	110395	46596	156956	7352	544	0.074	3957	0.538
Araucania	8105	91032	31930	91032	4349	332	0.076	2161	0.497
Lagos	6590	73535	27054	85040	4211	277	0.066	2259	0.536
Aysen	721	9330	3293	8886	461	33	0.072	232	0.503
Magallanes	671	9540	5113	16035	807	68	0.084	443	0.549
Metropolitana	20063	357793	200539	726155	34903	2275	0.065	19886	0.570
Rios	3240	38175	12505	34465	1757	137	0.078	838	0.477
Arica	919	13169	7566	24401	1251	80	0.064	767	0.613
Nuble	3349	43759	14461	41824	1928	152	0.079	1065	0.552
Overall	79600	1127548	536353	1847621	91717	6365	0.069	49947	0.545

2 Simultaneous Algorithm

We present the pseudocode for the *Simultaneous* heuristic. The algorithm takes as input a matching instance consisting of: a finite set of students S ; a finite set of schools C , where each school $c \in C$ offers $q_c^\ell \in \mathbb{Z}_+$ seats at level $\ell \in L$ (with $q_c^\ell = 0$ meaning school c does not offer level ℓ); a family partition $\mathcal{F} \subseteq 2^S$ of S , where $f(s) \in \mathcal{F}$ denotes the family of student $s \in S$; strict student preferences $(\succ_s)_{s \in S}$ over $C \cup \{\emptyset\}$; and initial school priority orders $(\succ_c)_{c \in C}$ over S , derived from group mappings $g(s, c)$ and random tie-breakers $(p_{s,c})_{s \in S, c \in C}$. The set of feasible student-school pairs is

$$V = \{(s, c) \in S \times C : c \succ_s \emptyset, q_c^{\ell(s)} > 0\},$$

where $\ell(s)$ denotes the level of student s .

Unlike the Descending benchmark, which processes levels sequentially from highest to lowest, the Simultaneous algorithm runs the student-proposing Deferred Acceptance (DA) algorithm over *all* students at once. After each run of DA, it updates the priority orders by promoting, to the top of the queue, any student s' whose sibling has just been matched to the same school. Specifically, if student s is assigned to school c , then for every sibling $s' \in f(s) \setminus \{s\}$ who also applies to c , student s' is elevated in c 's priority ranking. This update is *monotone*: once a student gains elevated priority at a school, that status is preserved in all subsequent iterations regardless of later changes to the matching. The process repeats until the matching stabilises, i.e., no student changes their assignment between two consecutive iterations.

Algorithm 1 Simultaneous

Input: Matching instance $(S, C, \mathcal{F}, (\succ_s)_{s \in S}, (\succ_c)_{c \in C}, (q_c^\ell)_{c \in C, \ell \in L})$

Output: Matching μ^t

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1: Set  $z_{s,c} \leftarrow 0$  for all  $s \in S, c \in C$  {no sibling priority initially}
2: Set  $\succ_c^0 \leftarrow \succ_c$  for all  $c \in C$  {start from initial priority orders}
3: Set  $t \leftarrow 0$ 
4: loop
5:   Compute  $\mu^t \leftarrow \text{DA}(S, (\succ_s)_{s \in S}, (\succ_c^t)_{c \in C}, (q_c^\ell)_{c \in C, \ell \in L})$ 
6:   if  $t \geq 1$  and  $\mu^t(s) = \mu^{t-1}(s)$  for all  $s \in S$  then
7:     break {matching has converged}
8:   end if
9:   for each  $s \in S$  such that  $\mu^t(s) \neq \emptyset$  do
10:    for each  $s' \in f(s) \setminus \{s\}$  do
11:      Set  $z_{s', \mu^t(s)} \leftarrow 1$  { $s'$  receives sibling priority at school  $\mu^t(s)$ }
12:    end for
13:  end for
14:  for each  $c \in C$  do
15:    Update  $\succ_c^{t+1}$ : students  $s$  with  $z_{s,c} = 1$  precede those with  $z_{s,c} = 0$ ; ties within each
    group are broken using  $\succ_c$ 
16:  end for
17:  Set  $t \leftarrow t + 1$ 
18: end loop
19: return  $\mu^t$ 
```

Notes.

- DA denotes the student-proposing Deferred Acceptance algorithm (Gale and Shapley 1962). Given school priority orders $(\succ_c^t)_{c \in C}$ and capacities (q_c^ℓ) , it returns the student-optimal stable matching with respect to those priorities.
- The variables $z_{s,c} \in \{0, 1\}$ track cumulative sibling priority: $z_{s,c} = 1$ means that at some iteration $\tau \leq t$ a sibling of s was matched to school c , and this status is never revoked. If sibling priorities are reset at the beginning of each iteration based on the current assignment, the algorithm may fail to converge, as students can repeatedly gain and lose priority at the same school across iterations.
- The priority update in lines 9–12 implements *absolute* contingent priority (Definition 3 in the paper): any student whose sibling has been matched to school c is placed in a strictly higher priority tier than students with no sibling matched to c , with ties within each tier broken by the initial order $\succ_c$.
- The algorithm terminates when two consecutive matchings coincide. Convergence is guaranteed in finite time because $|S|$ and $|C|$ are finite and the sibling-priority indicator

$z_{s,c}$ is monotone non-decreasing, so the set of priority-elevated students can only grow (or stay the same) between iterations.

Another algorithm that we can consider is DA + Improvement Cycles.